Qualcomm

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GKI Adoption Experience

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Agenda

1. GKI 1.0 Comments
2. Full GKI Commercialization Comments
3. Future of GKI @ Qualcomm Innovation Center
4. Out-of-Tree Module Challenges

GKI 1.0 Comments

v5.4 – 2020 Development Cycle

- Compliance testing with a GKI image
- Shipping devices with a diverged vendor kernel

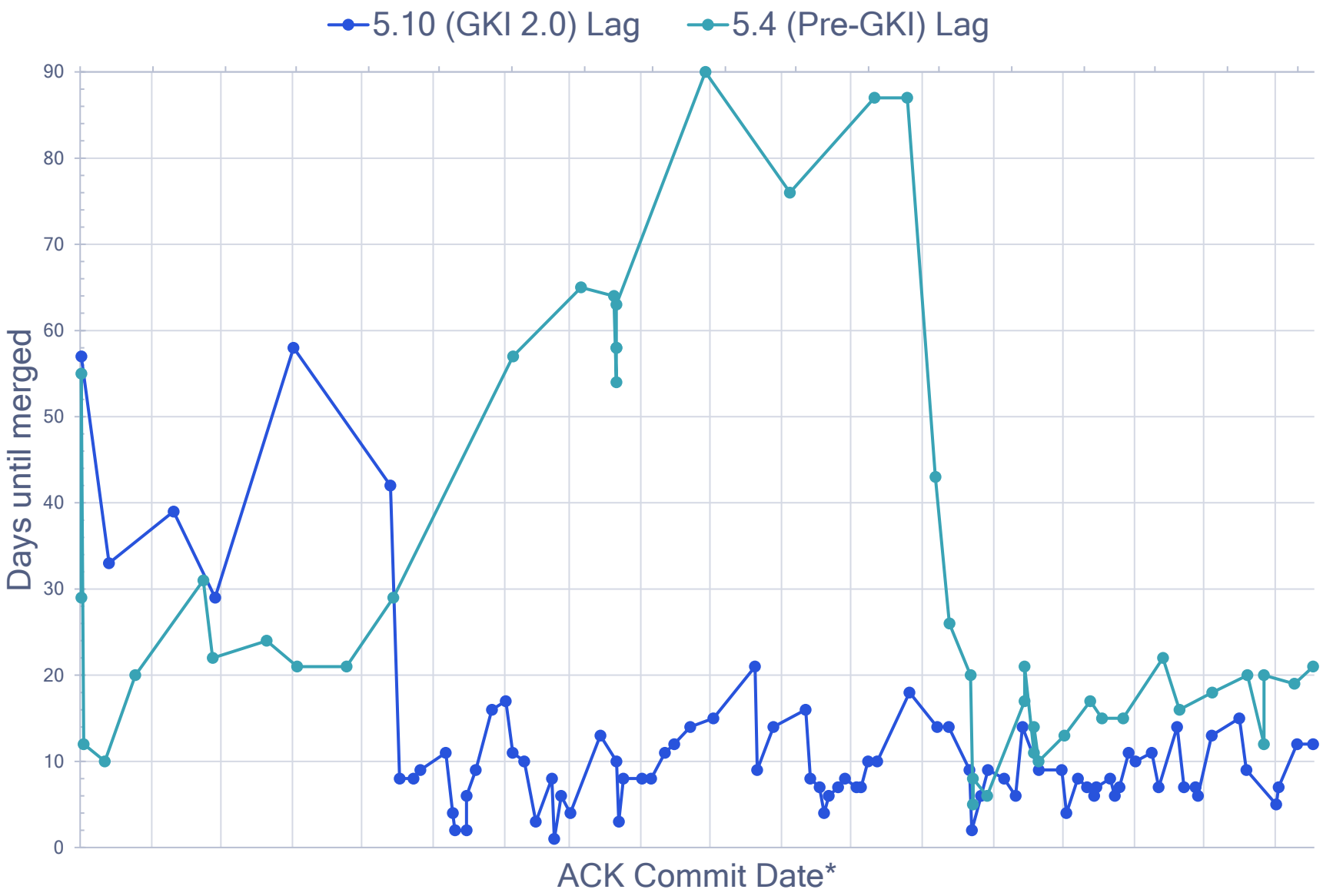
GKI 2.0 Comments

v5.10 - 2021

- Still using full kernel tree for vendor contributions
  - “Mixed build” – two full kernel trees: vmlinux from ACK; KOs from msm-kernel
- Embargoed changes challenges
  - Diverged kernel for approx. 3 months after launch which contain embargoed changes that we couldn’t release until device launches
  - Currently shipping Android Common Kernel GKI kernels for 5.10 and using public download boot.img*
- Security patches integrated quickly without any issues (e.g. Spectre-BHB)

* boot.img still needs to be signed by device vendor for Verified Boot

LTS Integration Lag



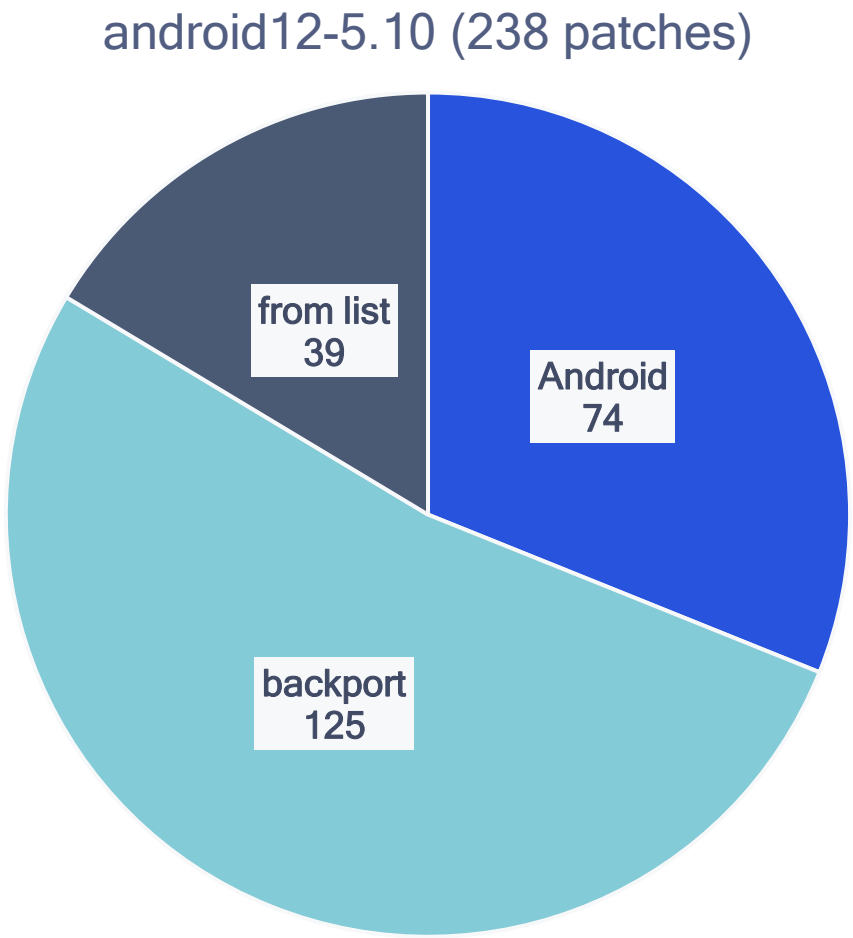
Increased LTS merge frequency!

Smaller lag between ACK and msm-5.10!

- 84 merges in 5.10 kernel
  - 46 in similar period for 5.4 kernel
- Average Delay: 8.3 days
  - 25.5 days for msm-5.4

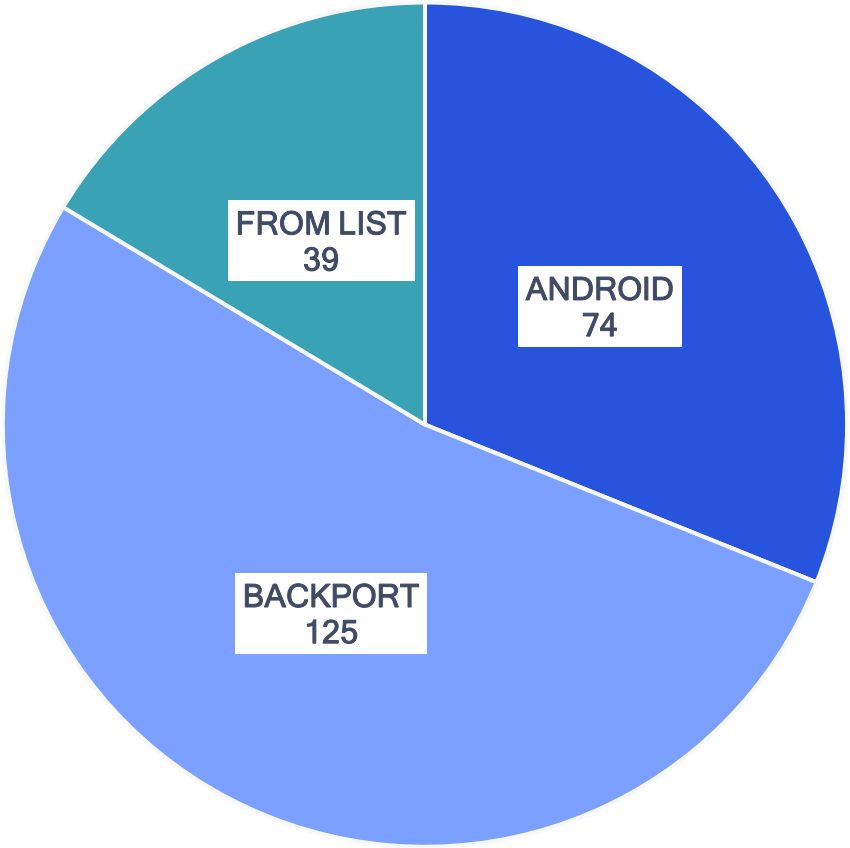
Our Upstream Contributions to Core Kernel

- ~125 contributions copied from kernel.org trees
  - 125 contributions would have otherwise been divergent changes carried in msm-5.10
  - Most active areas: USB, SCSI/UFS, iommu
- 39 contributions copied from mailing lists
  - Many ultimately accepted
  - “FROMLIST” happens when we urgently need change
- ~74 divergent changes
  - Excludes various GKI-specific changes such as vendor hooks, symbol exports, modifications to gki_defconfig
  - Most active areas: mm, scheduler

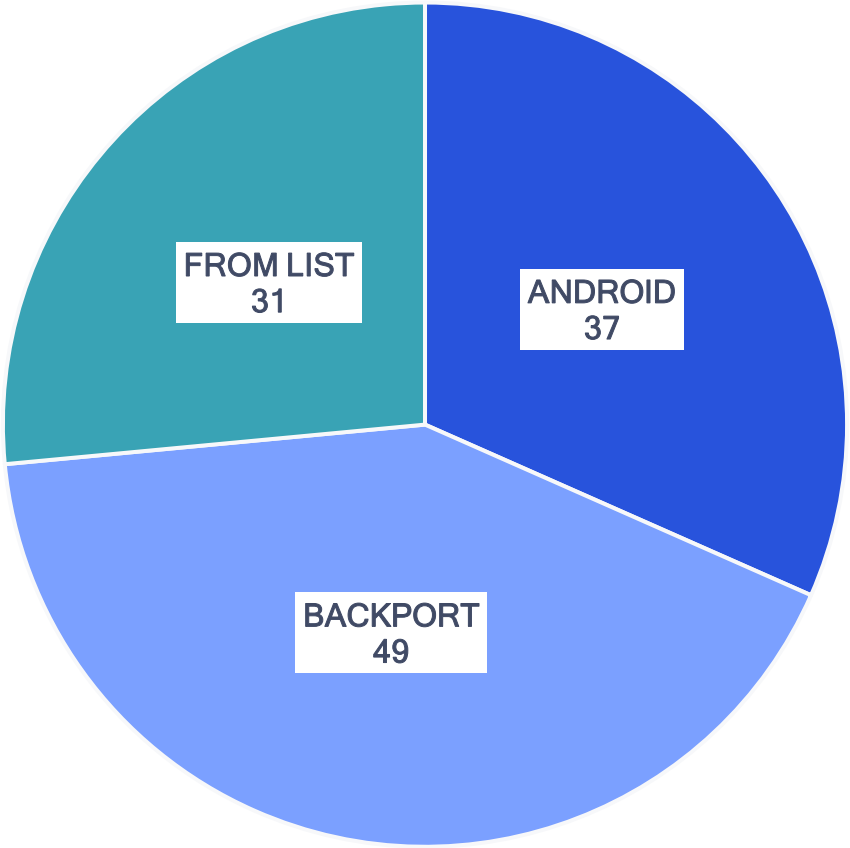


Our Upstream Contributions to Core Kernel

android12-5.10 (238 commits)



android13-5.15 (117 commits)



Future of GKI @ Qualcomm Innovation Center

- Exploring outside mobile use cases
- Challenges for those use cases:
  - GKI not yet mandated
  - Optimizing GKI for low memory devices
  - Maintaining boot performance indicators
  - Need to upstream first
- Exploring moving leaf drivers into an entirely out-of-tree project and consume Android Common Kernel without a complete vendor kernel tree

Out-of-Tree Module Challenges

- Kconfig
  - Working w/Google to solve this in Android Common Kernel
- Devicetree
  - Working w/Google to solve this in Android Common Kernel
- Output control
  - kernel src -> out of tree source
    has to be the same as:
    kernel out -> out of tree output
  - Solved independently by being careful with output trees

Thank you

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